

DERRICK MUTURI

Software Engineer | Machine Learning Engineer

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Software Engineer with 3+ years of experience building production web and mobile applications using React, TypeScript, Next.js, Node.js and PostgreSQL. Experienced in designing scalable REST APIs, secure authentication systems, database-backed applications and AI-enabled features. Hands-on experience developing machine learning models, Retrieval-Augmented Generation (RAG) applications, and rigorous model evaluation — including cross-validation, baseline benchmarking and data-integrity auditing — with a focus on delivering production-ready software.

SKILLS

Languages & Frameworks: JavaScript, TypeScript, Python, React, Next.js, React Native (Expo), Node.js, Express, HTML, CSS

Machine Learning: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, SVMs, Neural Networks, RAG, LLMs, cross-validation & model evaluation

Databases & Cloud: PostgreSQL, Supabase, MySQL, MongoDB, AWS, Microsoft Azure, Git/GitHub, Vercel

EXPERIENCE

Machine Learning Intern — Flyrank AI | Jul 2026 – Present

- Scoping and executing an applied ML research project — defining the research question, methodology and experimental direction.
- Building model experimentation workflows in Python notebooks, covering data handling, training and evaluation.
- Auditing and configuring AI-assisted development workflows and tooling to accelerate engineering productivity.

Frontend Developer — Africa Product Peers | Jun 2025 – Sep 2025

- Translated Figma handoffs into responsive, accessible and scalable React interfaces.
- Integrated RESTful APIs and built reusable UI components, improving UX consistency across product features.
- Contributed to performance optimisation and mobile responsiveness; shipped through sprint cycles with code reviews on Git/GitHub.

Full Stack Developer — PhysioMedix | Jun 2023 – Feb 2025

- Designed, developed and maintained scalable web applications for healthcare content delivery.
- Implemented authentication flows, role-based access control and security best practices across the platform.
- Designed and managed PostgreSQL database schemas via Supabase, handling complex queries and user data end-to-end.

PROJECTS

MACHINE LEARNING & GENERATIVE AI

XGauges — Football performance model with published evaluation • [Live](#) • [Code](#)

Python • scikit-learn • pandas • SciPy • FBref (soccerdata) • Fantasy Premier League API • React • Vite • Vercel

- Diagnosed target leakage in an earlier version of the model — the training label was arithmetic on its own input features — and rebuilt the project as two components: a transparent descriptive index ranking 177 Euro 2024 players within position via weighted z-scores, and a supervised forecast with a properly held-out label.
- Designed a leakage-proof evaluation using Euro 2024 metrics as features and actual 2025–26 Premier League output as labels, validated with 5-fold cross-validation, permutation importance and bootstrap confidence intervals.
- Benchmarked every model against mean and single-feature baselines and published the negative result — no trained model beat a minutes-only baseline (Spearman rho = 0.54, 95% CI [0.29, 0.72], n = 47) — rather than reporting an unvalidated accuracy claim.
- Identified survivorship bias in the label source, where a live API had already dropped relegated clubs and inflated model performance by roughly 10% MAE; added automated data-integrity checks and shipped a deployed React + Vite interface.

SongGrader — RAG-based sentiment analyser • [Live](#)

Python • LangChain • Spotify API • Genius API • OpenAI

- Built an end-to-end pipeline retrieving song metadata from Spotify and lyrics from Genius through REST API integrations.
- Developed a custom Retrieval-Augmented Generation workflow that evaluates lyrical sentiment against predefined scoring rubrics instead of relying solely on zero-shot prompting.
- Engineered prompt templates and retrieval logic to generate structured sentiment reports with consistent reasoning across songs.

MindPages — RAG-powered document Q&A • [Live](#)

Python • FastAPI • LangChain • ChromaDB • Hugging Face • OpenAI

- Built a Retrieval-Augmented Generation application allowing users to upload documents and query them in natural language.
- Implemented document parsing, recursive text chunking, embedding generation and semantic indexing with ChromaDB for efficient similarity search.
- Developed a retrieval pipeline injecting relevant document context into LLM prompts, producing grounded answers and reducing hallucinations.

SOFTWARE ENGINEERING

Stash — Bookmarking app that unifies saved social media content · [Live](#)

React Native · Expo · TypeScript · Node.js · PostgreSQL · Supabase

- Developed a cross-platform Android application centralising saved content from YouTube, Instagram, TikTok, Reddit and screenshots into searchable collections.
- Designed collections, tagging, notes, pinning and a Recap system that resurfaces previously saved content to encourage rediscovery.
- Built reusable React Native components and responsive interfaces focused on performance, intuitive navigation and a native mobile experience.

Resumify — AI-powered resume builder · [Live](#)

Next.js · React · Node.js · PostgreSQL · OpenAI API

- Developed a full-stack web application that generates tailored resumes using AI.
- Built user authentication, resume management and customisable templates.
- Integrated AI-powered content generation while keeping outputs fully editable by users.

EDUCATION & CERTIFICATIONS

- [BSc Data Science](#) — University of Nairobi
- [IBM AI Engineering Professional Certificate](#) — IBM (Coursera), Feb 2025
- [Software Engineering Certificate](#) — ALX Africa, Nairobi, Aug 2024
- [Full Stack Web Development](#) — Udemy, Nov 2024
- [MongoDB and the Document Model](#) — MongoDB University